1 Efficiency report

Overview

Menu	Production facility management → Key performance indicators → Efficiency report
Transaction code	effrp
Function authorization	effrp

The report includes workplace/machine-related performance data for a specific period of time and a specific number of workplaces. The result depends on the selections made and therefore on the selection criteria available in the selection panel. The performances regarding quantities and durations are displayed in a graphic. The production controller gets a quick and clear overview of the performances.

Selection criteria

The application provides the following selection criteria:

Workplace

This selection criterion refers to the workplace in the machine or workplace master data. You can also use wildcards (placeholders *).

Group

This selection criterion refers to the group in the machine or workplace master data. The application shows all machines or workplaces that are assigned to the selected group. You can also use wildcards.

Date from ... to ...

Fill in the fields from/to to narrow down the period to be evaluated

Shift / time

Restrict the defined period (date from/to). Select shifts or specify a time (from - until).

Report group

This selection criterion refers to the report groups. The application shows all machines or workplaces that are assigned to the selected report group.

Company

This selection criterion refers to the company defined in the machine or workplace master data. The application shows all machines or workplaces that are assigned to the selected company. You can also use wildcards.

Responsibility area

This selection criterion refers to the responsibility area in the machine master data. Note: The user can only view those machines that are included in the responsibility areas that are assigned to the user.

Short name

This selection criterion refers to the short designation of machines in the master data. The application shows all machines or workplaces matching the entered string. You can also use wildcards.

Cost center

This selection criterion refers to the cost center stored in the machine or in the workplace master data. All machines or workplaces are displayed that are assigned to the selected cost center. You can also use wildcards.

Designation

This field refers to the name of machines and workplaces defined in the machine master data. Only those machines are displayed that are identical to the entered string. You can also use wildcards (placeholders *).

Counter type

In the detail application "Consumption figures", you can select the counters to be displayed using the counter type as defined within the counter configuration of the machine. To use this detail application, you require the relevant license.

Selection options

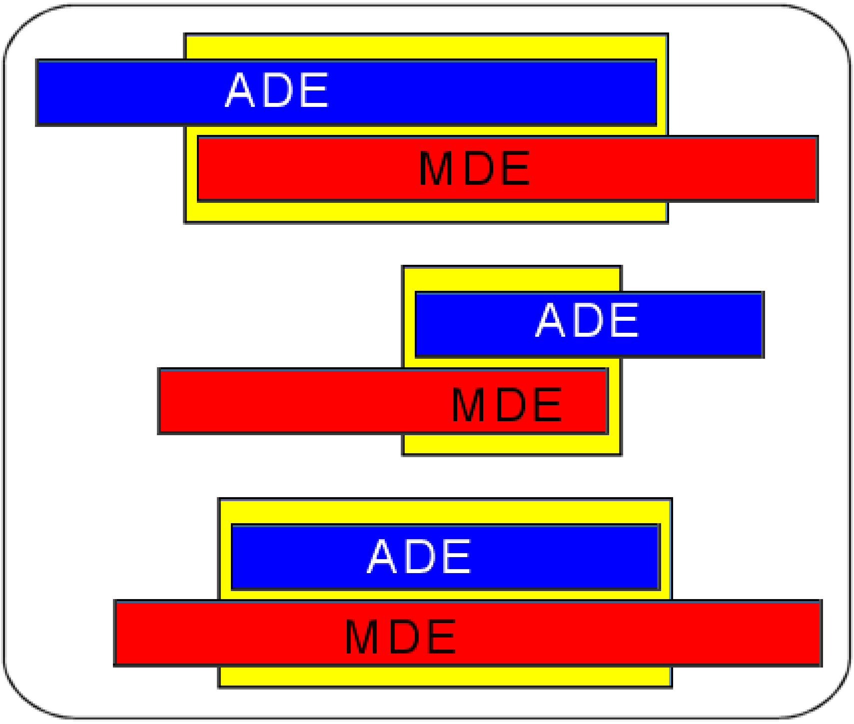
No selection:	If you enable this option, you cannot select the other selection criteria (order and resource).
Order:	You can use order parameters to restrict the data.
Resource:	You can use resource parameters to restrict the data.

Order/Final article/Batch number

If you select these options, the application only includes completed ADE postings. If the order is currently still running on the machine, the system does not take into account the time period between the last logon and now. Therefore it is possible that differences appear between the machine evaluation and the order-related evaluation. The application only includes ADE postings that coincide with the evaluation period. If necessary, choose a selection period making sure that the required ADE postings coincide with this period. For this order-related evaluation, MPDV recommends selecting data via the "shift" option instead of the "time".

The order number, article/item or the batch (from the order header) may be used as selection criteria.

The illustration below shows an example of how ADE and MDE postings can overlap. The ADE postings take priority in this evaluation type. The yellow areas show the result of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If several orders are logged on to the machine simultaneously, this evaluation type (select one of the options *order, final article* or *batch number)* assigns the full machine time (yellow area) and quantities to every order. The fact that orders are run in parallel will not result in a proportionate calculation.

Long-term data

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

Resource/Resource type

Restrict the selected posting records integrated in the evaluation by selecting the logged in resources or resources of a specific resource type.

Detail application *Efficiency report*

The detail application *Efficiency report* includes workplace/machine-related performance data for a specific period of time and a specific number of workplaces. The result depends on the selection and therefore on the selection criteria available in the selection panel.

Category *Workplace*

The following workplace/machine-related master data are available:

- Workplace
- Short name
- Designation
- Group
- Cost center
- Company

Category *Primary quantity, Secondary quantity, Tertiary quantity, Basic quantity*

Workplace/machine-related quantities recorded in the corresponding quantity types

- Yield
- Scrap
- Rework
- Open quantity

or the relevant quantity units (if relevant in the customer system).

Category *Durations*

- Production = RPA11

- Downtime = RPA01 + RPA02 + RPA03 + RPA04 + RPA05 + RPA06 + RPA07 + RPA08 + RPA09 + RPA10

- Total = production + downtime

Category *Cycles*

- Number of posted cycles

Category *Key figures*

- Rate of capacity utilization
  The rate of capacity utilization is the quotient deriving from effective run time and machine working time.
  Rate of capacity utilization = 100 / (RPA01 + RPA02 + RPA03 + RPA04 + RPA05 + RPA06 + RPA07 + RPA08 + RPA09 + RPA10 + RPA11) * RPA11
  Formula definition: rpa11 / (rpa1+rpa2+rpa3+rpa4+rpa5+rpa6+rpa7+rpa8+rpa9+rpa10+rpa11) * 100

  Use the formula **rcu** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- Assignment utilization rate:
  The assignment utilization rate represents the relationship deriving from the effective run time (main utilization time) and the adjusted machine operation time (i.e. without planned downtimes).
  Assignment utilization rate = 100 / (total of RPA 1 to RPA 11 minus RPA 6) * RPA 11
  Formula definition: rpa11 / (rpa1+rpa2+rpa3+rpa4+rpa5+rpa7+rpa8+rpa9+rpa10+rpa11) *100
  Use the formula **ocu** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- Efficiency (efficiency is only available when combined with the "lines" license)
  Efficiency is the quotient derived from the effective run time and the general run time, meaning the sum total of the effective run time and any interruptions as a result of machine-related disturbances or unscheduled shutdowns. Any other disturbances, e.g. organization-related disturbances, are not taken into account here.
  Efficiency = 100 / (RPA02 + RPA05 + RPA11) * RPA11

- Techn. efficiency
  The reference value for the technical efficiency is the sum total of the effective run time and interruptions because of technical (machine-related) disturbances. Other interruptions (e.g. of a organizational nature) are not considered: Techn. efficiency = 100 / (RPA02 + RPA11) * RPA11
  Formula definition: rpa11 / (rpa2 + rpa11) * 100
  Use the formula **tec_ef** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- Rate = 100 / ((yield (primary quantity) + scrap (primary quantity)) * rework (primary quantity) + open quantity (primary quantity)) * yield (primary quantity)
  Formula definition: yield.primary / (yield.primary + scrap.primary + rework.primary + problem.primary) * 100
  Use the formula **yie_ra** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

- Scrap rate = 100 / ((yield (primary quantity) + scrap (primary quantity) + rework (primary quantity) + open quantity (primary quantity)) * scrap (primary quantity)
  Formula definition: scrap.primary / (yield.primary + scrap.primary + rework.primary + problem.primary) * 100
  Use the formula **scr_ra** to customize the calculation. If the formula management does not include this formula, then you have to create the formula in order to change the calculation.

Note:
If you select the "time" option, the application only includes those MDE log records that are completely within the selected period (start and end must be within the selected period), that extend into the selected period (start or end is within the time frame) or that cover the complete selection period.

If you select the current shift where no MDE log records have been posted yet, the quantities produced within the selected period of time are calculated proportionally.

Example: At 10:00 am, you select an evaluation of the current shift from 8:00 to 9:00 am and the machine has the status production since 8:00 am. To calculate the recorded quantity in the selected period of time, the quantity produced since 8:00 (until 10:00) (1200 pieces) is divided according to the following formula:

((produced yield / complete duration of the current status in seconds) * evaluation duration)

In this example it is: (1200 / 7200) * 3600 = 600 pieces. The efficiency report will show a produced quantity of 600 pieces.

This proportionate calculation can have the effect that evaluated quantities can change within the current shift even if the selection parameters have not changed.

Detail application *Quantitative activities (machine-related)*

This detail application generates a graphic presentation of the quantities for the workplaces selected in the tabular detail application. Here, a differentiation is made by yield, scrap, rework and open quantity.

This detail application and the detail application *Quantity rate (group-related)* are docked one behind the other.

Detail application *Quantity rate (group-related)*

This chart shows, based on the selected entries in the table, the relationship between the quantities - yield, scrap, rework and open quantities. The information is displayed by group/ added total by workplace group.

This detail application and the detail application *Quantitative activities (machine-related)* are docked one behind the other.

Detail application *Durations*

This detail application generates a graphic illustration of the selected data from the efficiency report detail application. The durations are shown, broken down by RPA accounts (RPA01... RPA12).

You can hover the mouse over the graphic to display the value of the area where the mouse is. You can switch between displaying the value in percent or in duration.

Detail application *Consumption figures*

Shows the master data and counter values of the machine counters configured for the machines. By selecting the counter type, the data displayed can be limited to e.g. consumption figures or yield counters, or similar.

This detail application is only available if the system is configured accordingly and the relevant licenses are available.